Requirements and service boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete service-boundary record.

Keep the request, approved evidence, permitted operations, outcomes and unresolved work together.

Keep one habit: Recall → W1–W3/W5 → independent inventory → project contract → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

- The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Neural network experiment defense

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Timetable teaching service: approved public-schedule-classroom-v1 supplies Tuesday 16:00. A student asks to read Tuesday time and move the meeting to Wednesday. Read permitted and evidence present → answered with Tuesday16:00 and source. Move/edit excluded → out_of_scope, no change. Email and any other action are also excluded. An allowed read for Wednesday with no supplied entry → review, no guessed time.

  Review status is not a completed human decision. A malformed structured read missing day is invalid input. Separate museum picture: approved card P gives year 1912; read allowed, ticket sale excluded. Missing card Q makes the year unsupported/review; ticket sale remains excluded. Supplied source does not independently verify real history or current schedules.

  Local Python simulation takes structured operations and preserves the schedule; it has no live calls, authentication, natural-language parser, editor, email sender or actual reviewer. Test normal reads, mixed requests, absent entries and excluded actions with the same fixed contract.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Timetable teaching service: approved public-schedule-classroom-v1 supplies Tuesday 16:00. A student asks to read Tuesday time and move the meeting to Wednesday. Read permitted and evidence present → answered with Tuesday16:00 and source. Move/edit excluded → out_of_scope, no change. Email and any other action are also excluded. An allowed read for Wednesday with no supplied entry → review, no guessed time.

  Review status is not a completed human decision. A malformed structured read missing day is invalid input. Separate museum picture: approved card P gives year 1912; read allowed, ticket sale excluded. Missing card Q makes the year unsupported/review; ticket sale remains excluded. Supplied source does not independently verify real history or current schedules.

  Local Python simulation takes structured operations and preserves the schedule; it has no live calls, authentication, natural-language parser, editor, email sender or actual reviewer. Test normal reads, mixed requests, absent entries and excluded actions with the same fixed contract.

- A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Timetable teaching service: approved public-schedule-classroom-v1 supplies Tuesday 16:00. A student asks to read Tuesday time and move the meeting to Wednesday. Read permitted and evidence present → answered with Tuesday16:00 and source. Move/edit excluded → out_of_scope, no change. Email and any other action are also excluded. An allowed read for Wednesday with no supplied entry → review, no guessed time.

  Review status is not a completed human decision. A malformed structured read missing day is invalid input. Separate museum picture: approved card P gives year 1912; read allowed, ticket sale excluded. Missing card Q makes the year unsupported/review; ticket sale remains excluded. Supplied source does not independently verify real history or current schedules.

  Local Python simulation takes structured operations and preserves the schedule; it has no live calls, authentication, natural-language parser, editor, email sender or actual reviewer. Test normal reads, mixed requests, absent entries and excluded actions with the same fixed contract.

- A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Timetable teaching service: approved public-schedule-classroom-v1 supplies Tuesday 16:00. A student asks to read Tuesday time and move the meeting to Wednesday. Read permitted and evidence present → answered with Tuesday16:00 and source. Move/edit excluded → out_of_scope, no change. Email and any other action are also excluded. An allowed read for Wednesday with no supplied entry → review, no guessed time.

  Review status is not a completed human decision. A malformed structured read missing day is invalid input. Separate museum picture: approved card P gives year 1912; read allowed, ticket sale excluded. Missing card Q makes the year unsupported/review; ticket sale remains excluded. Supplied source does not independently verify real history or current schedules.

  Local Python simulation takes structured operations and preserves the schedule; it has no live calls, authentication, natural-language parser, editor, email sender or actual reviewer. Test normal reads, mixed requests, absent entries and excluded actions with the same fixed contract.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled requirements and service boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of requirements and service boundaries.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Requirements and service boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

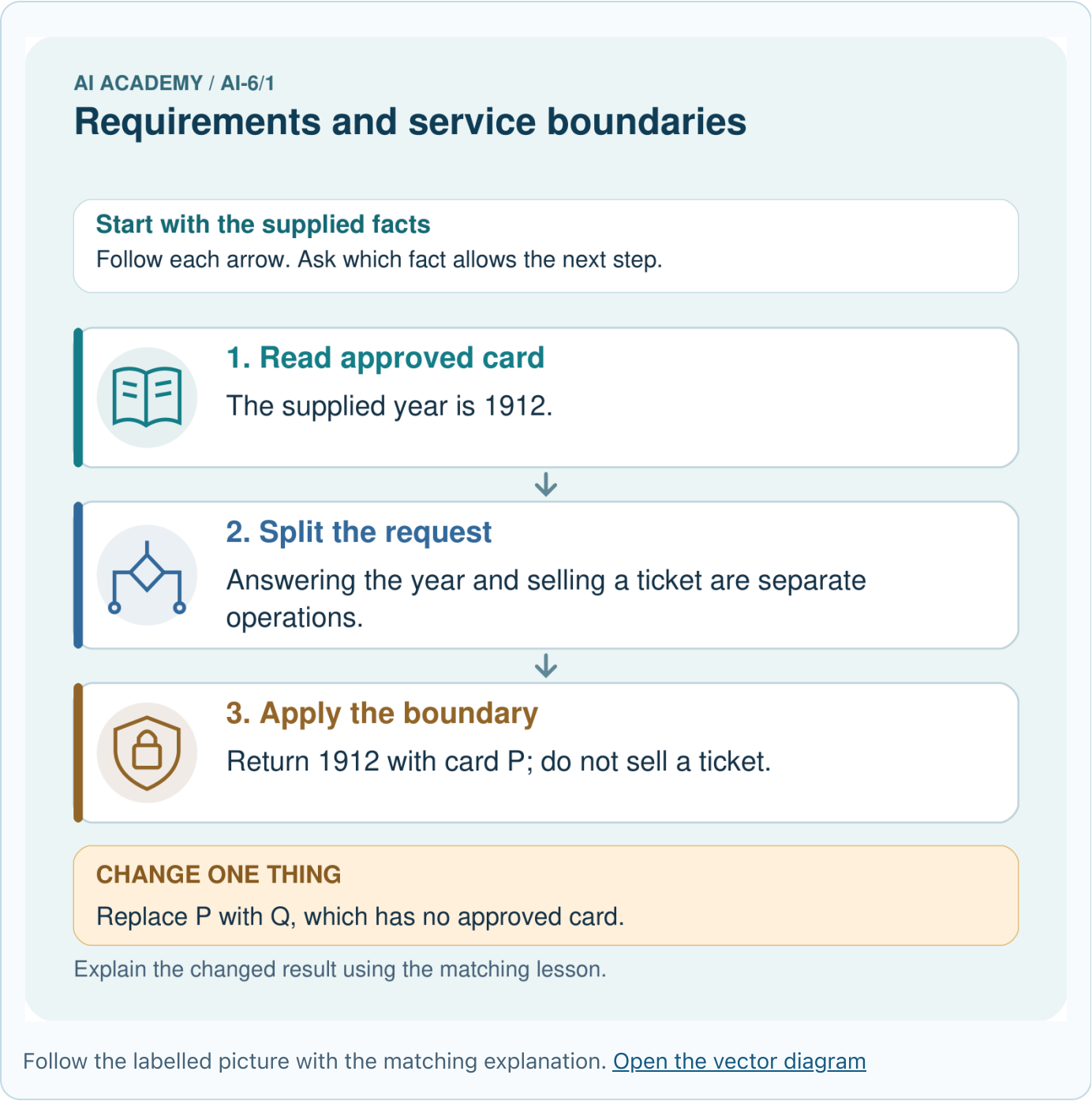
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.
- Approved card P says 1912. The contract permits reading cards, not selling tickets.
- Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.
- Which labelled step matches this detail? Return 1912 with card P; do not sell a ticket.
- 1. Read approved card
- 2. Split the request
- 3. Apply the boundary

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply the boundary Return 1912 with card P; do not sell a ticket. A **service boundary** states permitted work; useful information does not grant extra operations.

Changed-case question and optional hint

- Changed-case practice: Replace P with Q, which has no approved card. Which conclusion is supported by the supplied facts?
- 1. The year is unsupported, and the ticket sale is still excluded.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

Repair and retain the record

Review Neural network experiment defense (ai-5/36). Goal: Defend a network experiment using traceable **evidence**. Short practice: Calculate both weighted terms, z and ReLU **output**.

Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

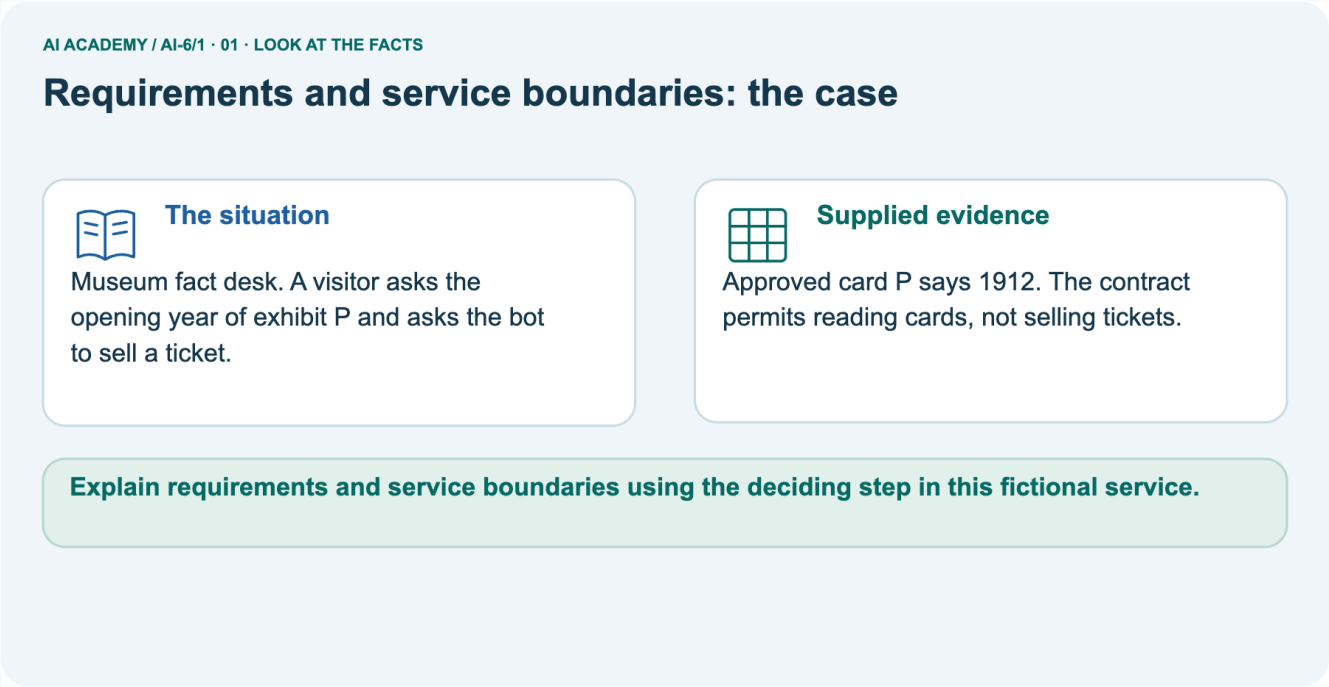
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Requirements and service boundaries: the case



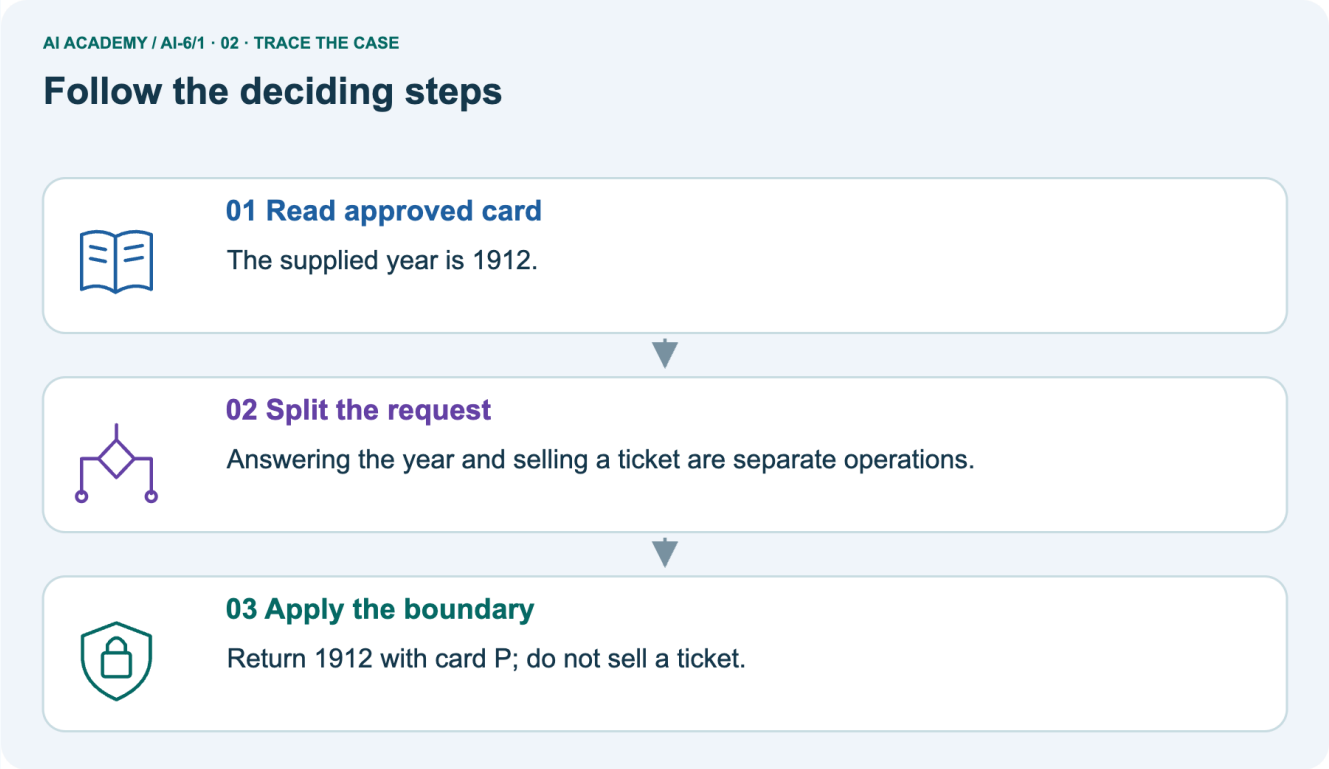
Follow the labelled picture with the matching explanation. Open the vector diagram

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket. Supplied facts: Approved card P says 1912. The contract permits reading cards, not selling tickets.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



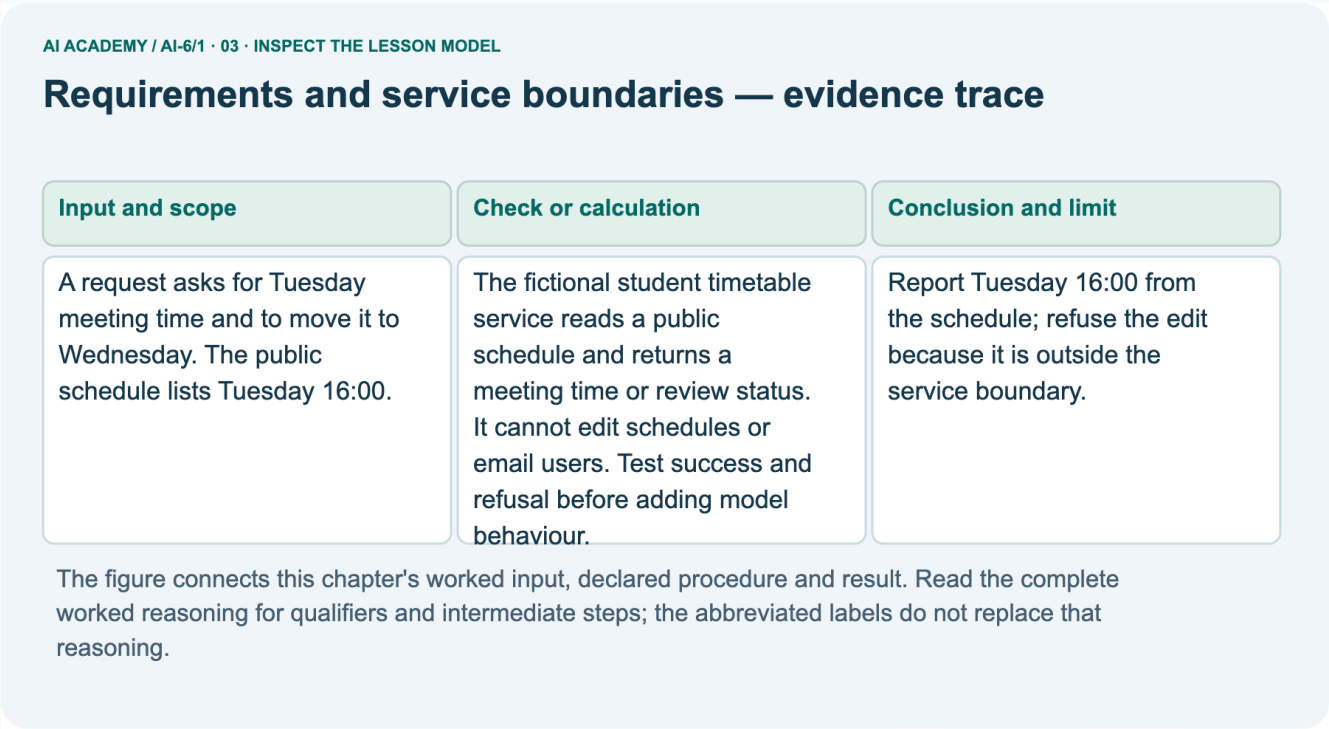
Follow the labelled picture with the matching explanation. Open the vector diagram

Read approved card: The supplied year is 1912. → Split the request: Answering the year and selling a ticket are separate operations. → Apply the boundary: Return 1912 with card P; do not sell a ticket.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Requirements and service boundaries — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements and service boundaries — **evidence** trace. A request asks for Tuesday meeting time and to move it to Wednesday.

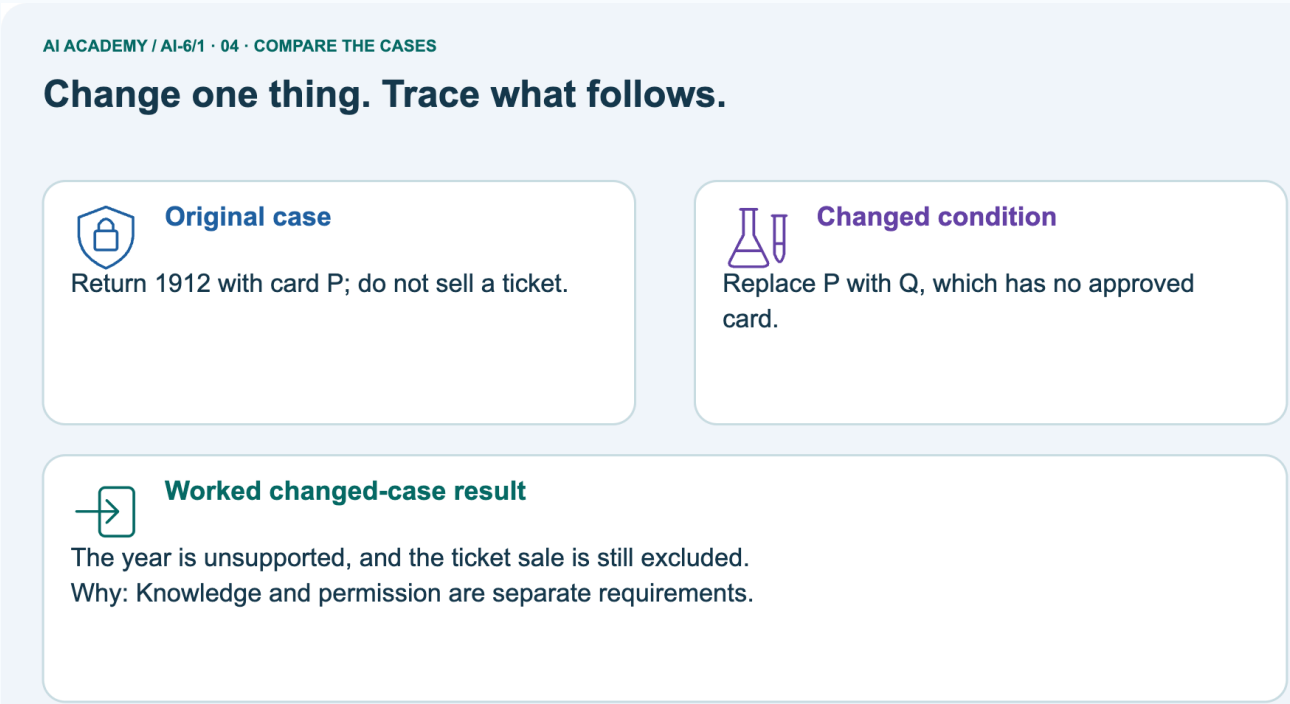
The public schedule lists Tuesday 16:00. | The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users.

Test success and refusal before adding model behaviour. | Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

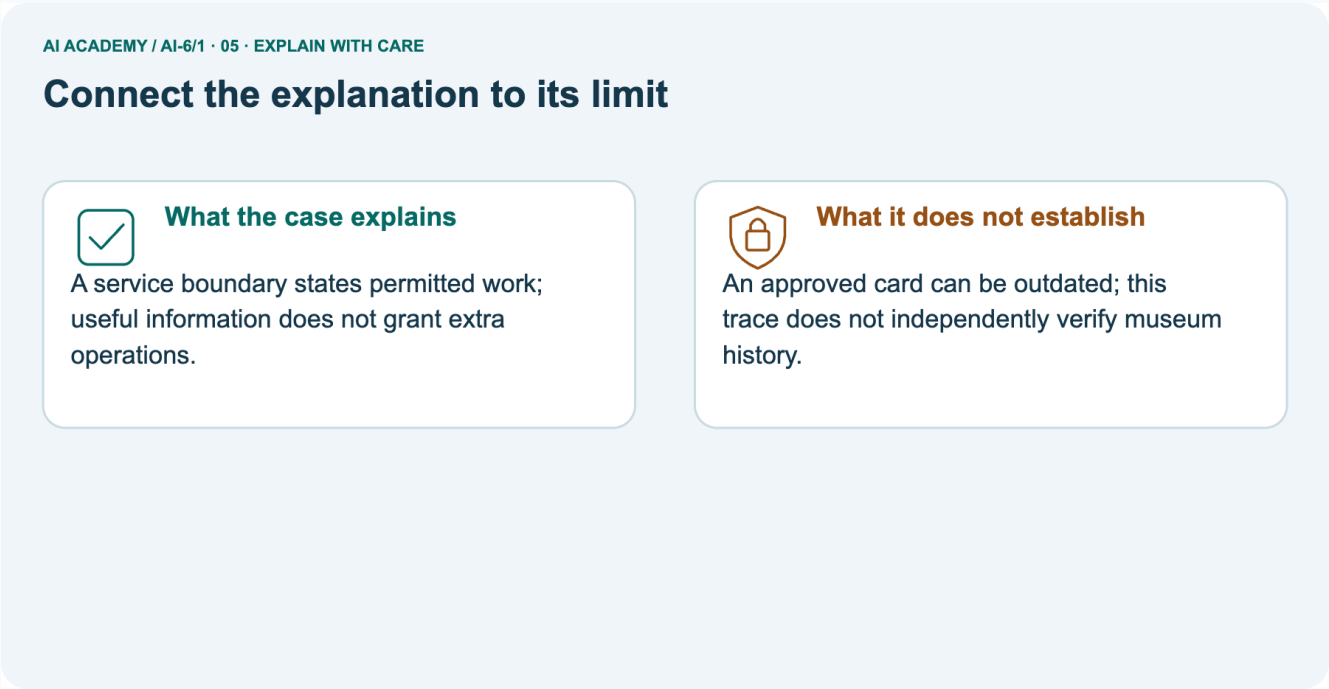
Original case: Return 1912 with card P; do not sell a ticket. Changed condition: Replace P with Q, which has no approved card. Expected result: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



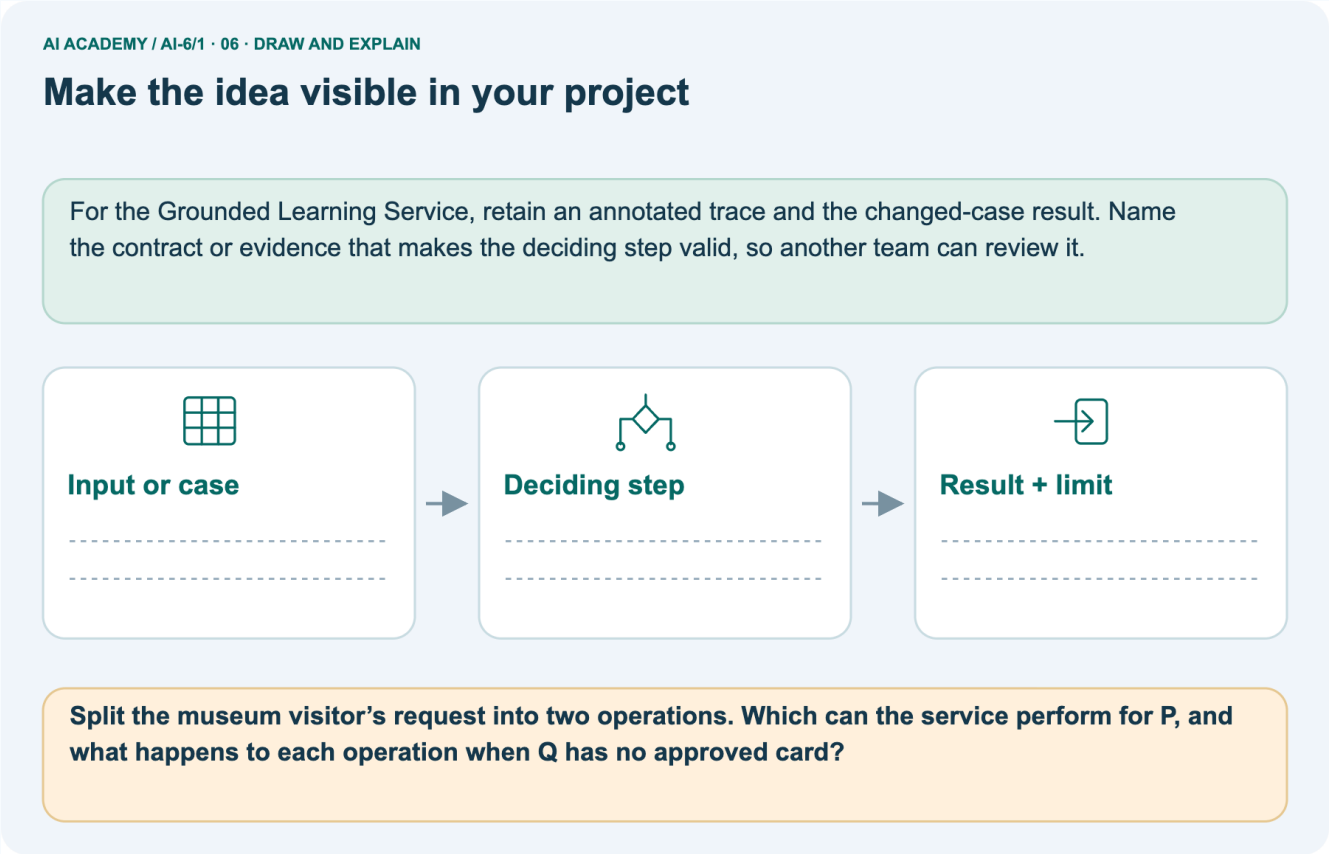
Follow the labelled picture with the matching explanation. Open the vector diagram

A **service boundary** states permitted work; useful information does not grant extra operations. Limit: An approved card can be outdated; this trace does not independently verify museum history.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Requirements and service boundaries: the case

02. Follow the deciding steps

03. Requirements and service boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.